

Machine Learning(CSAI 253) - Fall 2025

Phase 1

By

Team Members:

Abdallah Saed	202201933
Ahmed Mokhtar	202300820
Merna Ahmed	202201530
Mohamed Hamdy	202201599

Problem Statement

The main goal of this project is to figure out which employees are the most likely to leave the company, so we had to build a machine learning model to predict churn based on employee data such as workload, gender, salary, etc.

The dataset had **100,000 rows** and **35 features**

Data Exploration

The dataset had **100,000 rows** and **35 features** and it has 9.8% churn

And 90.2% not churn

Which means the data is very imbalanced, that caused some models to struggle.

I looked at distributions using histograms and boxplots. Some columns had big ranges, especially salary, satisfaction score, and working hours. There were a lot of outliers, but removing them deleted most of the dataset, so it wasn't realistic to use the "cleaned" version.

Correlation heatmaps showed almost no strong correlations with churn. Everything was near zero, which basically means the relationship with churn isn't linear. That's why tree-based models end up ruling this dataset.

Categorical features like department, marital status, and gender had noticeable differences between churners and non-churners.

Data Preprocessing

Missing Values

- Numerical → replaced with mean
- Categorical → replaced with mode

Inconsistencies

Made all categorical text lowercase and stripped spaces so models don't get confused by random formatting differences.

Feature Engineering

I added new features that actually describe employee behavior better:

- Overtime_ratio
- Projects_per_year
- Sick_days_per_year
- Training_per_year
- Promotions_per_year
- Satisfaction_performance
- Tenure
- CV_to_Hire_Gap

These help capture patterns that aren't obvious in the raw columns.

Binning

Created grouped categories for:

- Age
- Salary
- Performance
- Tenure
- Satisfaction

This helps models pick up thresholds instead of raw numbers.

Encoding & Scaling

- One-hot encoded all categorical columns

- Scaled numerical ones with StandardScaler
- Dropped datetime columns
The final dataset had **55 features**

Model Selection and Implementation

The project required 4 main classification models, so we used:

- KNN
- Logistic Regression
- SVM
- Decision Tree
- Random Forest
And then built a Voting Classifier as the ensemble technique.

Each model is useful in its way as:

- Logistic Regression is the baseline
- KNN struggles a bit with high dimensions
- SVM is powerful but sensitive to imbalance
- Decision Tree catches nonlinear relationships
- Random Forest handles the missy data best
- Voting tries to combine strengths of multiple models

Everything was trained using an 80/20 split so the class imbalance stays consistent.

Model Evaluation

Because this is classification problem , we used:

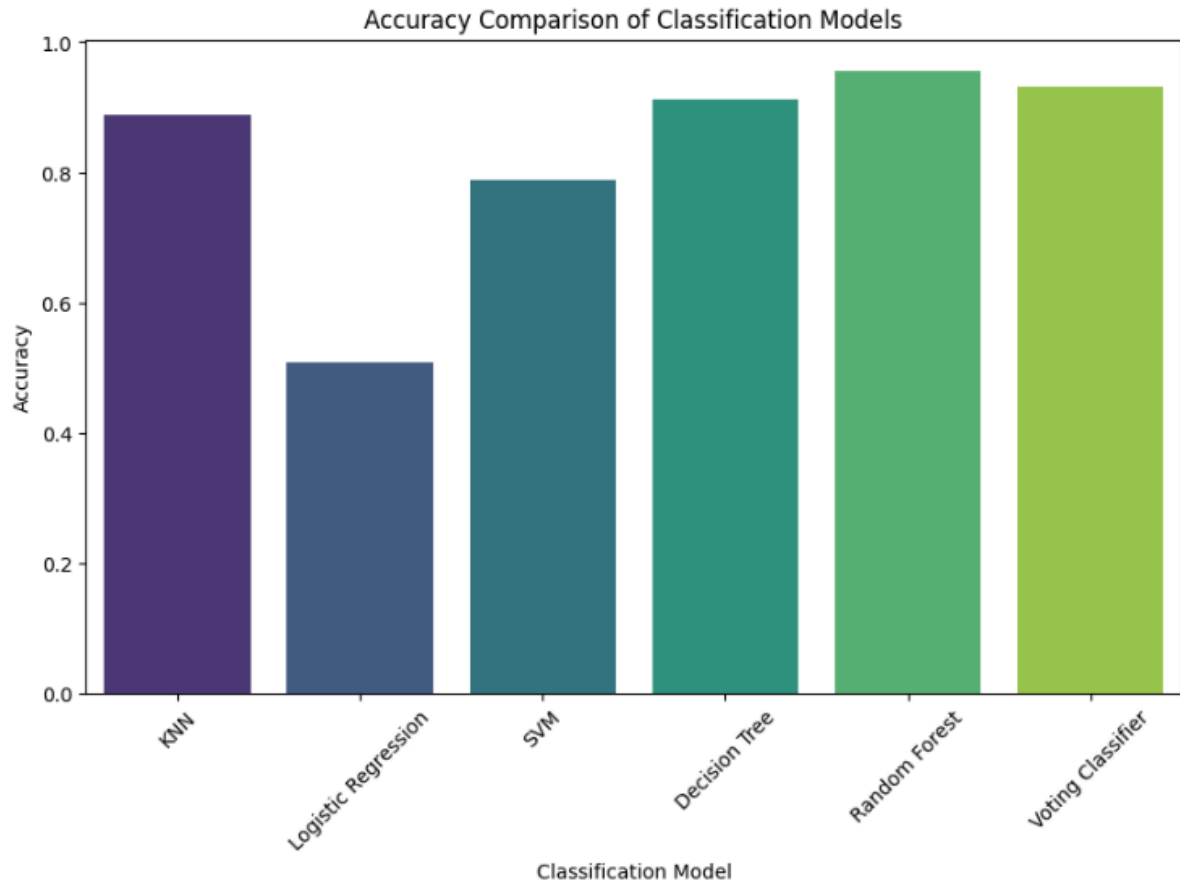
- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

	Model	Accuracy	Precision	Recall	F1 Score
0	KNN	0.88705	0.312583	0.119433	0.172830
1	Logistic Regression	0.50770	0.100345	0.500000	0.167146
2	SVM	0.78915	0.165223	0.279858	0.207778
3	Decision Tree	0.91225	0.550572	0.608806	0.578226
4	Random Forest	0.95590	1.000000	0.553644	0.712704
5	Voting Classifier	0.93105	0.791789	0.409919	0.540180

Insights

- Random Forest achieved the Highest accuracy, best F1, perfect precision.
- Voting Classifier improved results of weak models but still didn't achieve higher accuracy than Random Forest.
- Logistic Regression was the worst as the dataset is nonlinear and imbalanced.

Also the barplot below shows the accuracy comparison between models



Conclusion

This project showed that employee churn is not something linear models can easily understand. The data is imbalanced and the patterns are nonlinear, so the models that truly shine are the tree-based ones — especially Random Forest.

Models ranking based on performance:

1. **Random Forest**
2. Decision Tree
3. Voting Classifier
4. KNN
5. SVM

6. Logistic Regression

Limitations

- The strong class imbalance made recall for minority classes more challenging.
- Some engineered features had low correlation

Task Division:

Merna Ahmed:Random Forest implementation,ensemble technique, and report

Ahmed Mokhtar:EDA,KNN, and logistic regression models implementation

Abdullah Saed:Data loading,Cleaning,Feature engineering

Mohamed Hamdy:decision tree, SVM model implementation.